

# Learned Image Compression with Quantization Error Compensator

Nianxiang Fu<sup>1,2</sup>, Zhenzhong Chen<sup>1,2,\*</sup>, Huairui Wang<sup>3</sup>, and Shan Liu<sup>4</sup>

<sup>1</sup>Hubei LuoJia Laboratory, Wuhan University, Wuhan, China

<sup>2</sup>School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China

<sup>3</sup>Tencent, Shenzhen, China

<sup>4</sup>Tencent America, Palo Alto, CA, USA

**Abstract**—Recent advancements in learned image compression methods have demonstrated superior rate-distortion performance and remarkable potential compared to traditional compression techniques. However, the core operation of quantization, inherent to lossy image compression, introduces errors that can degrade the quality of the reconstructed image. To address this challenge, we propose a novel Quantization Error Compensator (QEC), which leverages spatial context within latent representations and hyperprior information to effectively mitigate the impact of quantization error. Moreover, we propose a tailored quantization error optimization training strategy to further improve rate-distortion performance. Notably, QEC serves as a lightweight, plug-and-play module, offering high flexibility and seamless integration into various learned image compression methods. Extensive experimental results consistently demonstrate significant coding efficiency improvements achievable by incorporating the proposed QEC into state-of-the-art methods, with a slight increase in runtime.

**Index Terms**—Learned image compression, quantization error compensator, quantization error.

## I. INTRODUCTION

Images, as a crucial form of digital media, pose challenges in storage and transmission due to their high dimensions. Traditional image compression standards like JPEG [1], JPEG2000 [2], and BPG [3] follow a transform, quantization, and entropy coding pipeline. Despite their success, these methods are limited by separately optimized frameworks and lack adaptability to diverse image contents.

Inspired by deep learning, recent approaches train neural networks to handle image compression end-to-end [4], [5], [6], [7], [8], [9], [10], [11], [12], often outperforming traditional methods. These learned methods typically replace traditional components with learnable modules. Specifically, a neural network transforms the image into compact latent representations, reducing redundancy. The continuous representations are then quantized into discrete counterparts for entropy coding, with an entropy model predicting their likelihood for efficient encoding. The decoding process reverses these steps.

While quantization is an effective method for reducing the bit rate of an image and is a necessary step in the compression process, it introduces quantization errors that can diminish the expressiveness of the latent representation. To

address this issue, Minnen *et al.* [7] proposed Latent Residual Prediction (LRP) to reduce quantization error. However, LRP exhibits performance degradation in zero-slice scenarios of the channel-wise autoregression (absence of autoregression). Fu *et al.* [13] introduced a Post-Quantization Filter (PQF) aiming to reduce the difference between original and quantized latent representations. Nonetheless, employing a single convolutional layer as a filter suffers from exploiting spatial contexts in latent representations fully, providing limited quantization error compensation.

In this paper, we introduce a Quantization Error Compensation module (QEC), utilizing spatial context within latent representations and hyperprior information to mitigate the impact of quantization. QEC aims to compensate for quantized latent representations, alleviating quantization distortion and bridging the gap between original and quantized latent representations. Additionally, we propose a tailored training strategy called quantization error optimization to further improve rate-distortion performance. With the cooperation of these two techniques, we can enhance current state-of-the-art methods while introducing acceptable time costs. To assess the efficacy of the proposed QEC, we integrate it into several state-of-the-art learned image compression methods and evaluate them on multiple datasets. Extensive experimental results consistently demonstrate the efficient enhancement of baseline models across various datasets. When incorporating QEC into Cheng2020 [8], it achieves an average BD-Rate savings [14] of -5.5% and -6.93% on the CLIC Professional Validation dataset [15] using MSE and MS-SSIM as distortion metrics, respectively. The proposed QEC is a highly flexible, plug-and-play module that is easily integrated into various learned image compression methods.

The remainder of the paper is organized as follows. Section II presents the proposed method in detail. Section III evaluates the proposed model through experimental results, performance comparisons with baselines, and an ablation study. Finally, the paper is concluded in Section IV.

## II. METHODOLOGY

### A. Architecture Overview

Fig. 1a illustrates the basic end-to-end image compression framework incorporating a hyperprior model [5] and a context model [6]. Specifically, inputting the image  $x$  into

\*Corresponding author: Zhenzhong Chen (Email: zzchen@whu.edu.cn). This work was supported by the National Natural Science Foundation of China under contract No. 62036005 and Tencent.

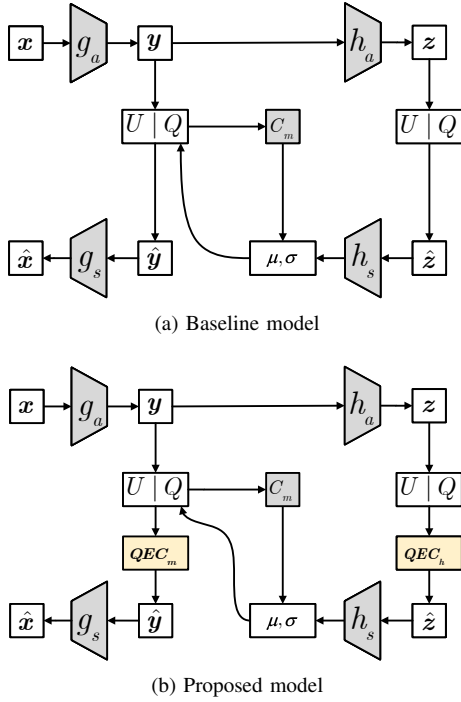


Fig. 1. Operational diagrams of learned image compression methods. (a) Baseline model. (b) Proposed model with quantization error compensator.

the analysis transform  $g_a$  produces latent representations  $\mathbf{y}$ . The representations are quantized to  $\hat{\mathbf{y}}$  using the quantization operation  $Q$ . Since  $Q$  is non-differentiable during training, it is approximated by adding uniform noise  $\mathcal{U}(-\frac{1}{2}, \frac{1}{2})$  to generate noisy latent representations  $\tilde{\mathbf{y}}$ . During inference,  $\mathbf{y}$  is rounded to  $\hat{\mathbf{y}}$  and encoded into a bitstream. Following [5], a hyperprior has been introduced as side information  $\mathbf{z}$  to capture the spatial dependencies in  $\mathbf{y}$  via the hyperprior encoder  $h_a$ , and  $\mathbf{z}$  is quantized in the same way as  $\mathbf{y}$ . A hyperprior decoder  $h_s$  and context model  $C_m$  are then used to estimate the Gaussian parameters  $\mu$  and  $\sigma$  of  $\hat{\mathbf{y}}$ , enabling the prediction of entropy of  $\hat{\mathbf{y}}$ . Finally,  $\hat{\mathbf{y}}$  is transformed into the reconstructed image  $\hat{\mathbf{x}}$  via the synthesis transform  $g_s$ .

Although quantization is an essential step for lossy compression, it introduces quantization error. Therefore, we place the proposed QEC module at the decoding side, serving as a feature-level post-processing module to mitigate the quantization error, as shown in Fig. 1b.

### B. Quantization Error Compensator

The quantization error is the difference between the latent representations and the quantized latent representations, given by:

$$\begin{aligned} \mathbf{r}_y &= \mathbf{y} - \hat{\mathbf{y}}, \\ \mathbf{r}_z &= \mathbf{z} - \hat{\mathbf{z}}, \end{aligned} \quad (1)$$

where  $\mathbf{r}_y$  denotes the main quantization error, while  $\mathbf{r}_z$  denotes the hyper quantization error.

The main quantization error compensator, denoted as  $QEC_m$ , addresses the main quantization error  $\mathbf{r}_y$  by taking  $\hat{\mathbf{y}}$  and the output of the hyper decoder (hyperprior) as inputs. Since the hyperprior is derived from  $\hat{\mathbf{z}}$ , it provides

supplementary information that aids in restoring  $\hat{\mathbf{y}}$ . These components are concatenated and passed through a convolutional layer for feature fusion, leveraging short-range spatial context. To fully utilize the information in  $\hat{\mathbf{y}}$ , we incorporate a Swin-Transformer-based attention module to capture long-range spatial context. The short-range and long-range spatial contexts are then aggregated. Finally, the process culminates in a convolutional layer with a Tanh activation function to produce the predicted quantization error, which is added to  $\hat{\mathbf{y}}$  to compensate for quantization loss, resulting in the corrected latent representations  $\bar{\mathbf{y}}$ .

Since  $\hat{\mathbf{z}}$  is part of the entropy model and is also affected by quantization, we compensate for  $\hat{\mathbf{z}}$  as well, improving the accuracy of the Gaussian parameters. The compensation process for  $\hat{\mathbf{z}}$  mirrors that of  $\hat{\mathbf{y}}$ , except it does not require additional priors as input.

The detailed architecture of the main quantization error compensator is illustrated in Fig. 2. In this figure, RBB denotes the residual bottleneck block shown on the right. The convolution parameters are denoted as (kernel size, number of filters, stride). Additionally, the SwinT Block refers to a Swin Transformer block [16].

### C. Quantization Error Optimization

The proposed QEC is optimized to predict quantization error using additional feature distortion loss functions,  $\mathcal{D}_f(\mathbf{y}, \bar{\mathbf{y}})$  and  $\mathcal{D}_f(\mathbf{z}, \bar{\mathbf{z}})$ , calculated as the MSE between  $\mathbf{y}$  and  $\bar{\mathbf{y}}$ , and  $\mathbf{z}$  and  $\bar{\mathbf{z}}$ , respectively.

To stabilize training, we use a two-stage approach. In the first stage, we incorporate the  $\mathcal{D}_f$  loss into the rate-distortion loss to form the total loss  $\mathcal{L}_t^1$ :

$$\begin{aligned} \mathcal{L}_t^1 &= \mathcal{R}(\hat{\mathbf{y}}) + \mathcal{R}(\hat{\mathbf{z}}) + \lambda \cdot \mathcal{D}(\mathbf{x}, \hat{\mathbf{x}}) + \beta \cdot \mathcal{D}_f(\mathbf{y}, \bar{\mathbf{y}}) \\ &\quad + \beta \cdot \mathcal{D}_f(\mathbf{z}, \bar{\mathbf{z}}), \end{aligned} \quad (2)$$

where  $\lambda$  controls the rate-distortion tradeoff;  $\mathcal{R}(\hat{\mathbf{y}})$  and  $\mathcal{R}(\hat{\mathbf{z}})$  represent the bitrates of  $\hat{\mathbf{y}}$  and  $\hat{\mathbf{z}}$ ;  $\mathcal{D}(\mathbf{x}, \hat{\mathbf{x}})$  denotes the distortion term; and  $\beta$  is a weight parameter. It is noteworthy that during the minimization of  $\mathcal{D}_f$ , if the overall framework is optimized end-to-end,  $g_a$  and QEC may take shortcuts by directly outputting latent representations  $\mathbf{y}$  that closely resemble  $\hat{\mathbf{y}}$ . This could lead to a subtle degradation in the expressiveness of  $\mathbf{y}$ , diminishing the performance benefits introduced by QEC. To solve this issue, we first detach  $\mathbf{y}$  and  $\hat{\mathbf{y}}$  from the computational graph and then input the detached latent representation into QEC. This enables the separate optimization of QEC without allowing its gradients to backpropagate into the main backbone of the network, thus preventing potential side effects. In the PyTorch framework, this functionality can be easily implemented using `y.detach()` and `y_hat.detach()`. The same approach is applicable to  $\mathbf{z}$ .

In the second stage, we fix the parameters of all modules except  $g_s$ ,  $h_s$ ,  $QEC_m$ , and  $QEC_h$ . The compensated latent representation is input into the synthesis networks, and we optimize these modules using  $\mathcal{L}_t^2$ :

$$\begin{aligned} \mathcal{L}_t^2 &= \mathcal{R}(\hat{\mathbf{y}}) + \mathcal{R}(\hat{\mathbf{z}}) + \lambda \cdot \mathcal{D}(\mathbf{x}, \hat{\mathbf{x}}) + \beta \cdot \mathcal{D}_f(\mathbf{y}, \bar{\mathbf{y}}) \\ &\quad + \beta \cdot \mathcal{D}_f(\mathbf{z}, \bar{\mathbf{z}}). \end{aligned} \quad (3)$$

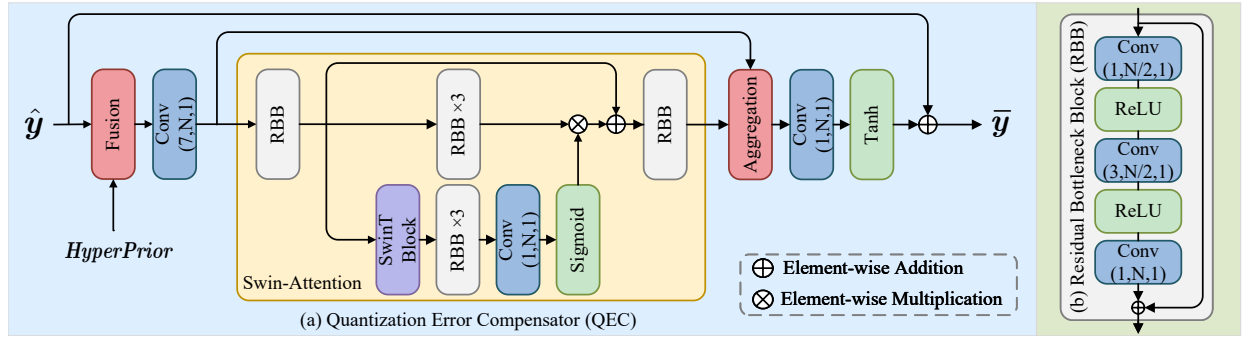


Fig. 2. The detailed architecture of the main quantization error compensator.

This strategy focuses on improving rate-distortion performance using the compensated latent representations.

### III. EXPERIMENTS

#### A. Experimental Setup

The proposed QEC is a plug-and-play module that's highly flexible and can be flexibly combined with other learned image compression methods. To demonstrate the effectiveness of the proposed QEC, we apply it to five baseline models: Factorized Prior [4], Scale Hyperprior [5], Mean-Scale Hyperprior [6], Mean-Scale Hyperprior with spatial context [6] (represented by "Mean-Scale Hyperprior + Context"), Cheng2020 [8], and TIC [10]. The selected baseline models capture dominant learned image compression architectures. Accordingly, the models enhanced by the QEC are represented by "Factorized Prior + QEC", "Scale Hyperprior + QEC", "Mean-Scale Hyperprior + QEC", "Mean-Scale Hyperprior + Context + QEC", "Cheng2020 + QEC", and "TIC + QEC", respectively. All models were implemented on the CompressAI [17] framework.

**Training.** We utilize Flickr2W [18] and select 50,000 images with size larger than  $256 \times 256$  from the validation set of ImageNet [19] as the training data, totaling approximately 70,000 images. During training, we perform random crops with a size of  $256 \times 256$ . To reduce training costs and time, we directly load the pre-trained weights from CompressAI and employ the Adam optimizer [20] with a batch size of 32 training all models for 14 epochs. The learning rate is set to  $1 \times 10^{-5}$ . In the initial 12 epochs, we optimize the entire framework using  $\mathcal{L}_t^1$ , while in the last 2 epochs, we optimize the modules  $g_s$ ,  $h_s$ ,  $QEC_m$ , and  $QEC_h$  using  $\mathcal{L}_t^2$ . Two quality metrics, namely MSE and MS-SSIM, are employed to evaluate distortion  $\mathcal{D}$ . When optimizing the model with MSE,  $\lambda$  is chosen from  $\{0.0018, 0.0032, 0.0075, 0.0150\}$ . For MS-SSIM optimization,  $\lambda$  is selected from  $\{2.40, 4.58, 8.73, 16.64\}$ . Additionally,  $\beta$  is set to 0.01 for both metrics. To ensure a fair comparison, baseline models without QEC are trained using the exact same strategy.

**Evaluation.** We evaluate our method on three datasets: Kodak dataset [21] with the image size of  $768 \times 512$ , Tecnick dataset [22] with the image size of  $1200 \times 1200$ , CLIC Professional Validation dataset [15] with 2K resolution. Both

PSNR and MS-SSIM are used to measure the distortion, while bits per pixel (bpp) are used to evaluate bitrate. MS-SSIM is converted into decibels, i.e.,  $-10 \log_{10}(1 - \text{MS-SSIM})$ .

#### B. Rate-Distortion Performance

Fig. 3 illustrates the rate-distortion performance of baseline models with and without the proposed QEC. Each point represents the average bits per pixel (bpp) and distortion for the test datasets. The comparison in Fig. 3 shows consistent improvements with QEC across various compression qualities and datasets in terms of both PSNR and MS-SSIM.

Table I presents the BD-Rate savings [14] for baseline models with QEC compared to their original versions on three datasets, using PSNR and MS-SSIM metrics. The results indicate significant BD-Rate improvements across all datasets and metrics, highlighting QEC's universal enhancement of model performance.

#### C. Ablation Study

To demonstrate the effectiveness of the proposed QEC and training strategy, we incorporated the QEC module into the decoders of Cheng2020 and Mean-Scale Hyperprior without using the proposed strategy for training (denoted as "Cheng2020 + QEC\*" and "Mean-Scale Hyperprior + QEC\*", respectively). As shown in Table II, using MSE as the distortion metric, the performance gain from employing QEC alone is significantly less than that achieved with our proposed loss function and training strategy. This also indicates that the use of QEC alone, as well as in conjunction with the proposed training strategy, contributes to performance enhancement.

#### D. Visual Quality

To demonstrate that QEC can mitigate the information loss caused by quantization, we present visual comparisons of reconstructed images in Fig. 4. It can be observed that incorporating QEC into Cheng2020 improves visual quality and preserves more details. The right side of Fig. 4 shows the quantization error feature maps of  $\mathbf{y}$  before and after compensation (mean of the absolute values of all channels of  $\mathbf{r}_{\mathbf{y}}$  and  $\bar{\mathbf{r}}_{\mathbf{y}}$ ). It is evident that the proposed QEC effectively reduces quantization errors.

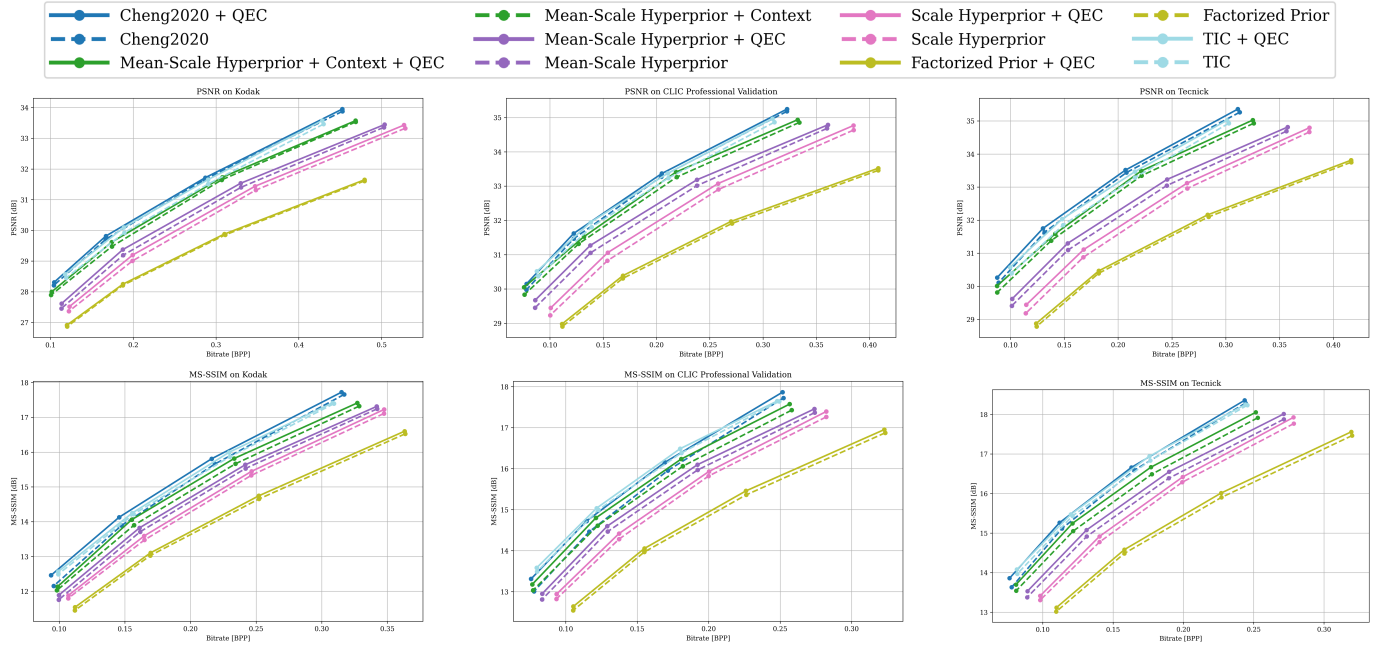


Fig. 3. The rate-distortion performance of baseline models and their enhanced versions by QEC in terms of PSNR and MS-SSIM on three datasets. The top row represents PSNR, while the bottom row corresponds to MS-SSIM.

TABLE I  
BD-RATE RESULTS AND AVERAGE DECODING TIME INCREASES OF BASELINE MODELS WITH QEC COMPARED TO THOSE WITHOUT QEC.

| Methods                                                  | Kodak  |         | CLIC   |         | Tecnick |         | Average Decoding Time Increase |
|----------------------------------------------------------|--------|---------|--------|---------|---------|---------|--------------------------------|
|                                                          | PSNR   | MS-SSIM | PSNR   | MS-SSIM | PSNR    | MS-SSIM |                                |
| Factorized Prior [4] (w. QEC vs. w/o QEC)                | -1.13% | -1.94%  | -2.08% | -2.36%  | -1.81%  | -2.47%  | 9.7%                           |
| Scale Hyperprior [5] (w. QEC vs. w/o QEC)                | -3.74% | -2.43%  | -4.79% | -2.98%  | -4.15%  | -3.06%  | 10.9%                          |
| Mean-Scale Hyperprior [6] (w. QEC vs. w/o QEC)           | -3.87% | -2.58%  | -5.13% | -3.42%  | -4.42%  | -3.85%  | 11.1%                          |
| Mean-Scale Hyperprior + Context [6] (w. QEC vs. w/o QEC) | -1.52% | -3.76%  | -3.78% | -5.44%  | -3.41%  | -4.89%  | 2.5%                           |
| Cheng2020 [8] (w. QEC vs. w/o QEC)                       | -3.22% | -5.36%  | -5.50% | -6.93%  | -3.23%  | -4.08%  | 3.5%                           |
| TIC [10] (w. QEC vs. w/o QEC)                            | -4.03% | -2.0%   | -4.28% | -2.49%  | -4.56%  | -2.54%  | 2.7%                           |

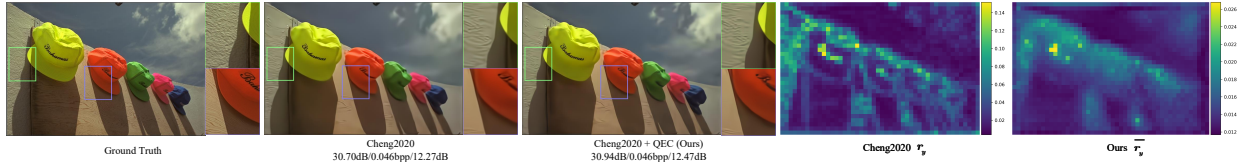


Fig. 4. Comparisons of the reconstructed images and quantization error feature maps of *Kodim3* from Kodak with Cheng2020 as the baseline model. Values are PSNR/bpp/MS-SSIM, respectively.

TABLE II  
BD-RATE RESULTS OF BASELINE MODELS WITH QEC USING THE PROPOSED QUANTIZATION ERROR OPTIMIZATION AND WITHOUT IT (DENOTED AS “w/o QEO”) COMPARED TO THOSE WITHOUT QEC.

| Methods                            | Kodak  | CLIC   | Tecnick |
|------------------------------------|--------|--------|---------|
| Mean-Scale Hyperprior w/o QEO      | -1.19% | -0.55% | -0.43%  |
| Mean-Scale Hyperprior + QEC (Ours) | -3.87% | -5.13% | -4.42%  |
| Cheng2020 w/o QEO                  | -1.39% | -2.86% | -1.33%  |
| Cheng2020 + QEC (Ours)             | -3.22% | -5.50% | -3.23%  |

### E. Computational Complexity

Table I shows the average decoding time increase on the CLIC Professional Validation dataset (2K resolution) when applying QEC to various baseline models. The decoding time increase is modest, ranging from 2.5% to 11.1%. Additionally,

using Cheng2020 as a baseline, the parameter count increases from 29.6M to 31.4M with the addition of QEC, representing a 6.1% increase. This demonstrates that QEC is lightweight.

## IV. CONCLUSIONS

We propose a Quantization Error Compensator (QEC) module to enhance learned image compression by leveraging spatial context and hyperprior information to mitigate quantization distortion. Additionally, we introduce a tailored quantization error optimization training strategy to further improve rate-distortion performance. QEC is a lightweight, plug-and-play module that integrates easily into various learned image compression methods. Experimental results demonstrate QEC’s effectiveness across various codecs and multiple datasets.

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